<u>Last time:</u> $f: A \to B$ is...

① surjective if: $\forall b \in B$, $\exists a \in A$ s.t. $f(a) = b$.

② injective if: $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$

③ bijective if: both ① and ②

<u>Exc</u> If $f: A \to B$ is bijective, then $\forall b \in B$ $\exists$ exactly one $a \in A$ s.t. $f(a) = b$.

<u>Def</u> Let $f: A \to B$ be a bijective function. The <u>inverse</u> of f is the function $f^{-1}: B \to A$ given by $f^{-1}(b) = a$, where $a \in A$ is the unique element in A s.t. $f(a) = b$.

<u>Rmk</u> If f is bijective, then $f^{-1}(f(a)) = a$ $\forall a \in A$ and $f(f^{-1}(b)) = b$ $\forall b \in B$.

<u>Ex</u> $f(x) = 2x$

① $f: \mathbb{R} \to \mathbb{R}$ is bijective.

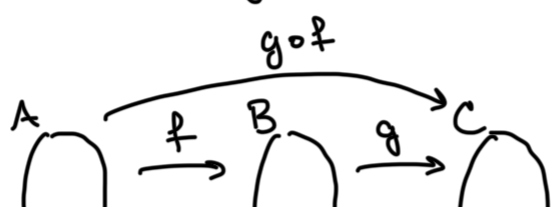
$f^{-1}: \mathbb{R} \to \mathbb{R}$, $f^{-1}(x) = \frac{1}{2}x$ is a function.

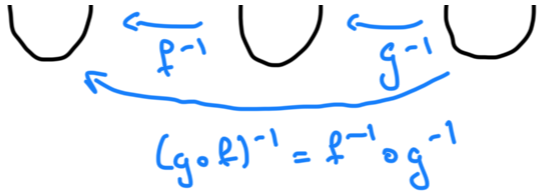
② $f: \mathbb{N} \to \mathbb{N}$ is not bijective.

$f^{-1}(x) = \frac{1}{2}x$ is not a function $\mathbb{N} \to \mathbb{N}$

(e.g. $1 \mapsto \frac{1}{2} \notin \mathbb{N}$)

<u>Exc</u> If $f: A \to B$ and $g: B \to C$ are bijective, then $g \circ f: A \to C$ is bijective and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.





CARDINALITY (Rudin Ch. 2)

<u>Def</u> We say that 2 sets A and B have the same <u>cardinality</u> if $\exists$ a bijection $f: A \to B$. When this is true, we write $A \sim B$.

<u>Def</u> Let A be a set. We say:

① A is <u>finite</u> if:

- $A = \emptyset$ (in which case A has <u>cardinality 0</u>), or
- $A \sim \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$ (in which case A has <u>cardinality n</u>).

② A is <u>countable</u> if $A \sim \mathbb{N}$.

③ A is <u>at most countable</u> if A is finite or countable.

④ A is <u>uncountable</u> if A is not at most countable.

<u>Warning</u>: Some authors use "countable" for ③ as well. By our (and Rudin's) convention, countable sets are infinite.

<u>Rmk</u> If A is countable, then we can enumerate the elements of A. Indeed, if $f: A \to \mathbb{N}$ is bijective, then

$$A = \{a_1, a_2, a_3, \dots\}.$$

$a_1 \mapsto 1$
$a_2 \mapsto 2$
$a_3 \mapsto 3$
$\vdots$

$f^{-1}(1) \quad f^{-1}(2) \quad \dots$

<u>Ex</u> The set $A = \{x, y, z\}$ has cardinality 3, since $f: A \to \{1,2,3\}$ is bijective.

$$x \mapsto 1$$
$$y \mapsto 2$$
$$z \mapsto 3$$

<u>Ex</u> Prove that $B = \{2n : n \in \mathbb{N}\} = \{2, 4, 6, \dots\}$ is countable.

$g(n)$: 1 2 3 ...

<u>Pf</u>: Set $g: B \to \mathbb{N}$, $g(m) = \frac{m}{2}$.

Claim: g is injective. Suppose $g(m_1) = g(m_2)$. Then $\frac{m_1}{2} = \frac{m_2}{2}$, and so $m_1 = m_2$.

Claim: g is surjective. Fix $k \in \mathbb{N}$. Then $2k \in B$ and $g(2k) = \frac{2k}{2} = k$.

Together, we conclude g is bijective. Therefore B is countable. $\square$

- In general, it's a fact that:

<u>Exc</u> If A is countable and B is an infinite subset of A, then B is countable.

- See Thm 2.8 in the book for details

<u>Prop</u> The set of integers

$$\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-n : n \in \mathbb{N}\}$$

is countable.

Idea:

$\mathbb{Z}$:	$\dots$	-2	-1	0	1	2	$\dots$
$f(n)$:		4	2	1	3	5	

Pf: Set $f: \mathbb{Z} \to \mathbb{N}$,

$$f(n) = \begin{cases} 2n+1 & n \geq 0 \\ -2n & n < 0 \end{cases}$$

Want: f is bijective. By HW1#5, it suffices to find an inverse $g: \mathbb{N} \to \mathbb{Z}$ for f. Set

$$g(m) = \begin{cases} \frac{m-1}{2} & m \text{ odd} \\ -\frac{m}{2} & m \text{ even} \end{cases}$$

Then

$$(f \circ g)(m) = \begin{cases} f(\overbrace{\tfrac{m-1}{2}}^{\geq 0}) = 2 \cdot \frac{m-1}{2} + 1 = m & \text{if } m \text{ odd} \\ f(\underbrace{-\tfrac{m}{2}}_{<0}) = -2 \cdot (-\tfrac{m}{2}) = m & \text{if } m \text{ even} \end{cases}$$

$$(g \circ f)(n) = \begin{cases} g(\overbrace{2n+1}^{\text{odd}}) = \frac{(2n+1)-1}{2} = n & \text{if } n \geq 0 \\ g(\underbrace{-2n}_{\text{even}}) = -\frac{(-2n)}{2} = n & \text{if } n < 0 \end{cases}$$

for any $m \in \mathbb{N}$, $n \in \mathbb{Z}$. Therefore, f is bijective by HW1#6, and so $\mathbb{Z}$ is countable. $\square$

Rmk Notice the steps we followed:

1. Come up with a pattern for enumerating the elements of A.
2. Write down a formula for $f: A \to \mathbb{N}$.

③ Prove f is bijective.